OM BHARADWAJ

bharadwajom07@gmail.com | LinkedIn | GitHub | LeetCode | +91 6207803739 | Jamshedpur, India (open to relocation)

SUMMARY

Computer Science undergraduate at IIIT Manipur (CGPA 9.05) who builds medical imaging models and turns them into working apps. Ocula, a diabetic retinopathy screening platform, is the basis of my first-author paper accepted at IEEE SILCON. Cutis screens skin lesions on ISIC 2024 and adds explainable predictions and RAG reports grounded in clinical guidelines. At Tata Technologies I worked on an enterprise CRM portal. Looking for software engineering, backend or applied AI roles.

TECHNICAL SKILLS

Languages: Python, JavaScript, C++, Java, SQL, PHP, C
Backend & Web: FastAPI, Node.js, Express.js, PHP, REST APIs, Microservices, JWT auth, role-based access control, React.js
Databases: PostgreSQL, MySQL, MongoDB, SQLite
Machine Learning: PyTorch, scikit-learn, XGBoost, LightGBM, CatBoost, Random Forest, SVM, ADASYN, Optuna, Pandas, NumPy
Models: EfficientNet-B4, ResNet-18, DenseNet-121, GoogLeNet, ConvNeXt, Swin-T, EdgeNeXt, EVA-02
Explainability, LLMs & RAG: Grad-CAM, Score-CAM, LIME, SHAP, Groq LLaMA-3, Gemini API, hybrid BM25 + embedding search
DevOps & Testing: Docker, Docker Compose, Nginx, Git, GitHub, CI/CD, Render, automated testing
Fundamentals: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks, System Design

EXPERIENCE

Tata Technologies May 2026 to July 2026
Software Engineering Intern *Jamshedpur, Jharkhand, India*

- Worked in an agile team on an enterprise CRM portal that large clients use to manage their customer relationships.
- Built a responsive React.js frontend with dashboards, charts and real-time notifications.
- Wrote REST APIs in PHP with role-based access control, authentication middleware and automated reporting.
- Designed MySQL schemas with normalized tables, indexed queries and stored procedures so the system could scale.

PROJECTS

Ocula: Diabetic Retinopathy Screening Platform Demo Video | GitHub
Python, PyTorch, FastAPI, Node.js/Express, React, PostgreSQL, Docker

- Built a full-stack screening platform. On APTOS 2019 it reaches 98.50% binary accuracy (AUC 0.997) and 90.68% five-class accuracy, 0.70 and 1.39 points above the Butt et al. (2022) baseline. It is the basis of my first-author paper accepted at IEEE SILCON.
- Ensembled five CNN backbones (ResNet-18, GoogLeNet, EfficientNet-B4, DenseNet-121, ConvNeXt-Tiny) with 8-view test-time augmentation and soft-voting classical classifiers (SVM, RF, XGBoost). Binary accuracy rose 1.38 points over the best single backbone.
- Designed a two-level hierarchical SVM cascade with ADASYN oversampling for five-class grading. It gained 1.27 points over a flat classifier and lifted recall on the minority Severe DR class from 0.79 to 0.82.
- Added Grad-CAM, LIME and SHAP explanations plus two Groq LLaMA-3 reports (clinical and patient). A full prediction with explanations takes under 45 seconds on CPU.
- Deployed as four Docker Compose services (React, Node.js/Express, FastAPI, PostgreSQL) with JWT auth and role-based access, on a validated clinical dataset of 3,662 images.

Cutis: AI Skin Lesion Screening Platform Demo Video | GitHub
Python, PyTorch, FastAPI, Groq LLaMA-3, Node.js/Express, React, MySQL, Docker

- Built a full-stack skin lesion screening platform and trained its ISIC 2024 ensemble on the 400K+ image SLICE-3D dataset (401,059 lesions, 393 malignant, and an always benign model scores 99.90% accuracy). On the challenge's official pAUC80 metric it reaches 0.1683, with 0.9653 AUROC, out-of-fold under 5-fold cross-validation grouped by patient, flagging 32.1 lesions per cancer found at 80% sensitivity and ranking 80.8% of each patient's cancers in their top 15 lesions. The ISIC 2024 winner scored 0.1726 pAUC80, 0.9668 AUROC and 0.790 top 15 coverage on the hidden test set. Probabilities are calibrated to Brier 0.00084 and ECE 0.00005.
- Deployed as four Docker Compose services (React, Node.js/Express, FastAPI, MySQL) with JWT auth and patient and clinician roles that change screens, scores, library access and reports. Grad-CAM (with a Score-CAM fallback), LIME and SHAP explanations run in parallel on every scan with quality metrics. A Groq LLaMA-3 agent writes reports and follow-up answers for each role, grounded in an NCI PDQ and CDC guideline library through hybrid BM25 + embedding search, with citations checked on the server. The app runs the image branch only, since one photo lacks the inputs the tabular stage needs, and scores a photo in about 4 seconds on CPU.
- Ensembled 20 CNNs (EVA-02-S, EdgeNeXt-B, ConvNeXt-T and Swin-T, 5 folds each) with test-time augmentation on two views and z-score standardization against stored out-of-fold statistics, then fused them with a tabular GBDT stage (203 features) and an Optuna-tuned LightGBM head. This raises pAUC80 from 0.1646 (tabular only) to 0.1683, and leave-one-institution-out evaluation holds at 0.1647. One-vs-rest AUROC is 0.9517 for Melanoma, 0.9770 for BCC and 0.9771 for SCC, with the type call 73.3% accurate among true cancers. Training takes 3.6 hours on two Tesla T4 GPUs, and all 20 networks serve in about 2 GB of RAM.
- Verified with 106 automated tests (50 Node, 46 Python including 32 for retrieval, 10 web) against a real MySQL server, and a replay tool that checks the app's scoring against stored out-of-fold predictions on 300 lesions (Spearman > 0.8).

EDUCATION

Indian Institute of Information Technology Manipur Imphal, India
B.Tech, Computer Science & Engineering (AI & Data Science) *2023 to 2027 (Expected)*
CGPA: 9.05 / 10.0 (Top 5% of the cohort)

ACHIEVEMENTS & LEADERSHIP

- Top 5% of my cohort at IIIT Manipur across five semesters, with a CGPA of 9.05.
- GDG On Campus Chapter Lead, IIIT Manipur. Selected through Google's competitive process. I run Study Jams, workshops and developer events for 1200+ students.
- GeeksforGeeks Campus Ambassador. I help students find technical resources and get started with competitive programming, and I solve problems regularly on LeetCode.